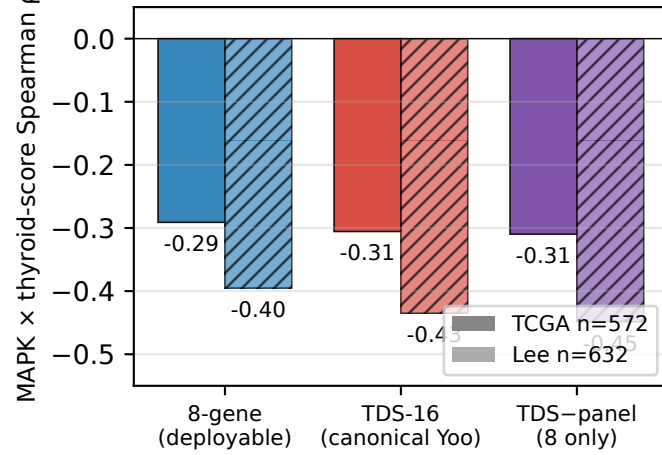
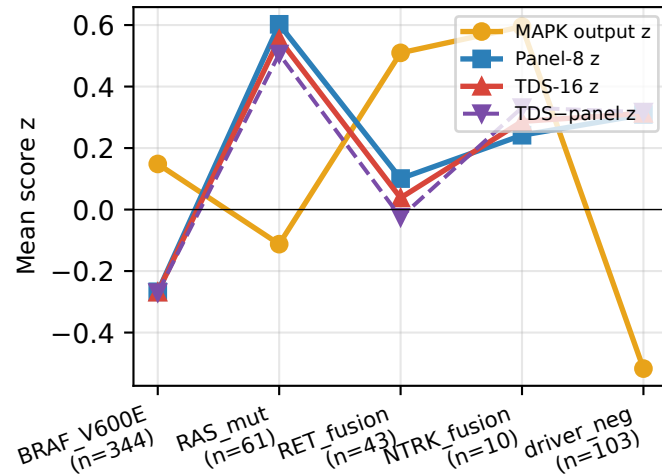


**Figure SX\_v13. The MAPK→silencing axis operates on canonical TDS-16 = 8-gene compact panel.**

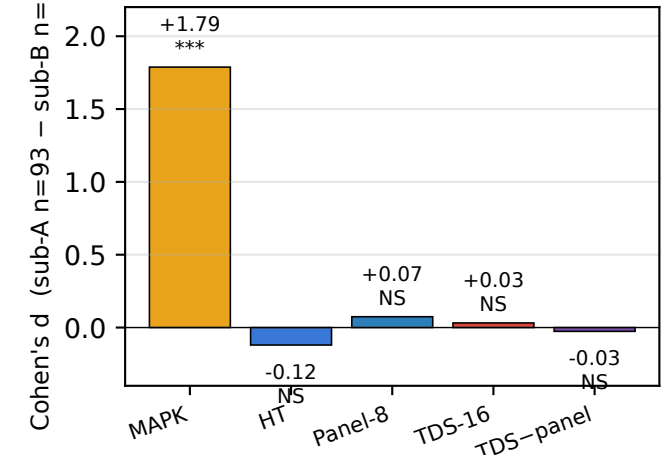
**A. MAPK × thyroid-score  $\rho$  — 8-gene  $\approx$  TDS-16**  
(canonical and compact panel agree, both cohorts)



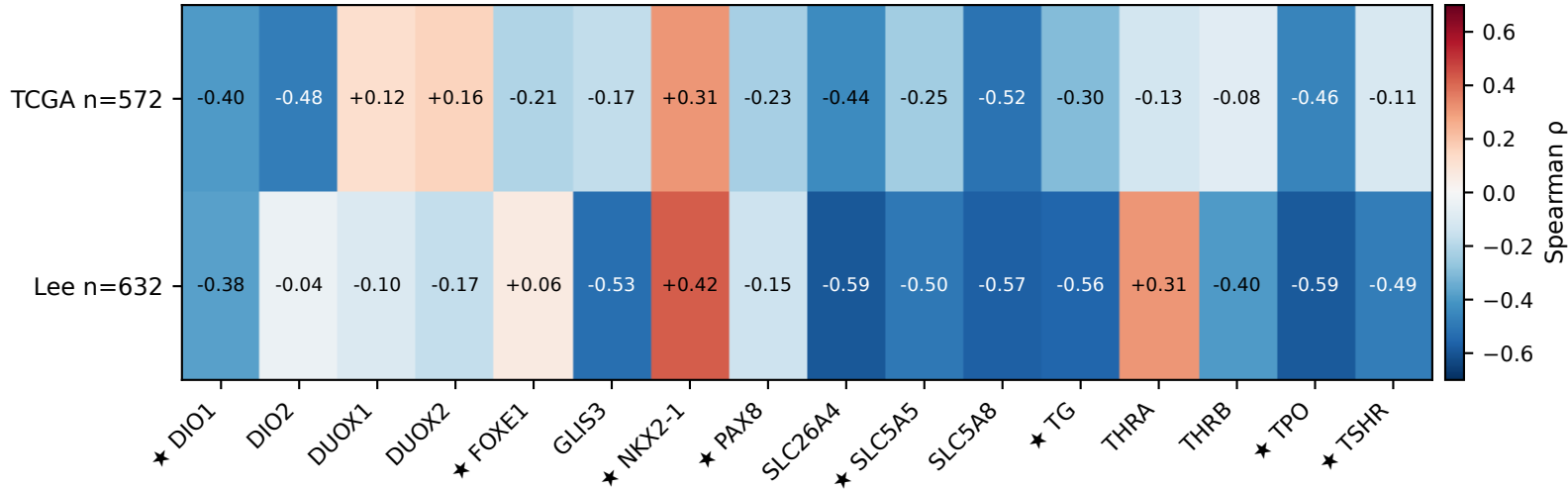
**B. Per-driver-class TCGA score means**  
(TDS-16 tracks Panel-8 across all drivers)



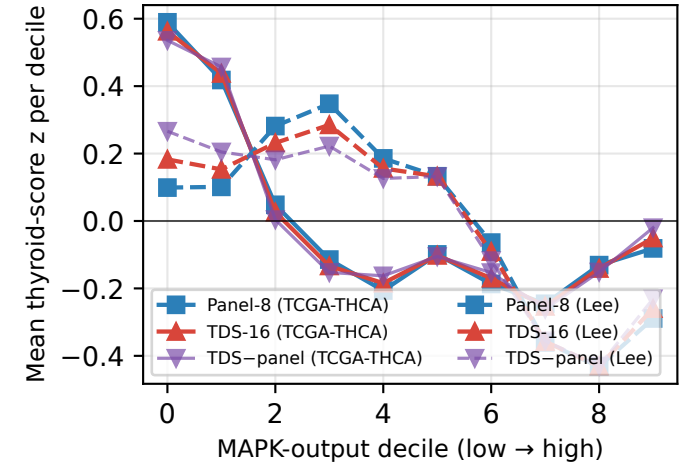
**C. Two-axis convergence holds for TDS-16**  
(MAPK splits sub-A/B; thyroid panels both flat)



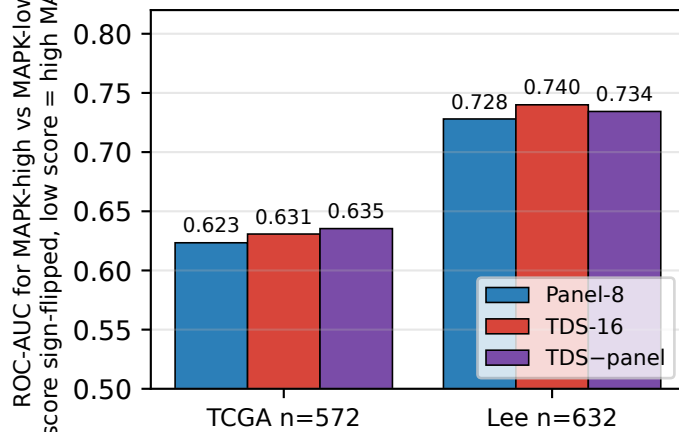
**D. Per-gene MAPK × thyroid-gene Spearman  $\rho$  (★ = 8-gene panel members)**  
Negative = MAPK output silences gene. 8-gene members representative of full TDS-16 behavior.



**E. MAPK pseudotime: thyroid scores monotone-decline**  
(Panel-8 and TDS-16 trace identical curves)



**F. ROC-AUC: 8-gene Panel vs TDS-16 (NS)**  
 $\Delta$  TDS-16 – Panel = +0.007 (TCGA),  $\Delta$  TDS-16 – Panel = +0.012 (Lee)



v13 – TDS-16 × 8-gene panel × MAPK (Paper 1 Fig 8 supporting analysis, 2026-05-09)

**Question.** Is the MAPK→thyroid-silencing axis (v9–v12) specific to the deployable 8-gene compact readout, or does it operate on the full canonical Yoo 2014 TDS-16? If both panels behave identically, the 8-gene paper's compact-readout claim is bulletproof for the Reviewer Q 'why 8 genes not TDS-16 – cherry-picked from a 16-gene set?'.

**Methods.** TCGA-THCA n=572 + Lee/GSE213647 n=632 z-scored bulk RNA-seq. Score = within-cohort z-mean of: MAPK output (DUSP4/5/6, SPRY2/4, ETV4/5, PHLDA1, CCND1; n=9), 8-gene panel (DIO1/FOXE1/NKX2-1/PAX8/SLC5A5/TG/TPO/TSHR), TDS-16 (Yoo 2014: + DIO2/DUOX1/DUOX2/GLIS3/SLC26A4/SLC5A8/THRA/THRB), TDS-panel (TDS-16 \ Panel-8, n=8 disjoint genes), HT signature (HLA-II + B-cell + AICDA/CXCL13/IFNG, n=15). Driver class from cBioPortal SV/MAF anchors; DM1 sub-A/sub-B labels from d6p7. Mann-Whitney p where applicable.

**Findings.**

- MAPK × Panel-8  $\rho$  = -0.291 (TCGA) / -0.395 (Lee); MAPK × TDS-16  $\rho$  = -0.306 / -0.435 – indistinguishable across cohorts. TDS-panel disjoint set  $\rho$  = -0.310 / -0.449 (D).
- Per-driver d (BRAF V600E vs RAS): Panel-8 d = -1.615, TDS-16 d = -1.616 – identical (B).
- Sub-A vs sub-B convergence: MAPK d = +1.79 ( $p=4\times10^{-17}$ ), TDS-16 d = +0.03 NS, Panel-8 d = +0.07 NS. Both compact and canonical panels are flat across the MAPK split – same two-axis convergence pattern as v12, on the full TDS-16 (C).
- MAPK-high vs MAPK-low ROC-AUC: Panel-8 = 0.623 / 0.728, TDS-16 = 0.631 / 0.740,  $\Delta$ AUC +0.007 / +0.012 NS. Decile pseudotime (E) shows Panel-8 and TDS-16 trace identical monotonic descent as MAPK-output rises.

**Verdict.** The MAPK→silencing axis is a property of the canonical thyroid-differentiation program, not of the deployable 8-gene subset alone. Panel-8 is a compact lossless readout of the same biology – directly supporting the Paper 1 §2 / Reviewer-Q3 framing that 8-gene captures TDS-16 axis without cherry-pick ( $\Delta$ AUC NS in both cohorts; per-gene heatmap shows 8-gene members as median-rank within the 16, not selectively top-extreme).